APT-BENCH: PATIENT-DISJOINT CELL TYPING FROM SINGLE-CELL APTAMER PROFILES

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ABSTRACT

Single-cell aptamer profiles offer a targeted representation for immune-cell typing, but evaluating additional cells from known patients does not establish generalization to new patients. We introduce APT-Bench, a benchmark specification and reference evaluation of 361,792 cells from 40 participants, with 293 aptamer measurements and RNA-derived labels for five lineages and 27 subtypes. The benchmark combines five fixed patient-disjoint folds, subject-balanced pooled Macro-F1, and classical and neural reference models. Across the available XGBoost and Soft-cascade Transformer split controls, cell-level splitting yields higher coarse and fine Macro-F1 in all four comparisons, by 0.050–0.108 on the cell-weighted metric. These comparisons demonstrate protocol-dependent optimism, rather than isolating a causal leakage effect. Under the patient-disjoint contract, reference models achieve coarse/fine subject-balanced Macro-F1 of 0.298–0.335/0.126–0.145, establishing a challenging evaluation setting for subsequent APT models. APT-Bench makes the unit of generalization explicit when assessing this measurement modality; complete data access and portable reproduction remain requirements for its public release.

1 INTRODUCTION

Single-cell aptamer profiling measures targeted binding patterns alongside transcriptomic annotations. Apt-seq established joint aptamer and transcriptome sequencing (Delley et al., 2018); the measurement principle is not new. The question here is whether these profiles support immune-cell typing in patients absent from training. General single-cell benchmarks provide reusable task definitions (Luecken et al., 2025), but gene-expression inputs do not directly define an evaluation contract for a fixed aptamer panel. A modality-specific reference task must distinguish additional cells from additional independent patients.

Clinical single-cell collections can contain hundreds of thousands of cells from only tens of people. APT profiles may encode both cell identity and patient-associated shifts; cell-random evaluation can reward the latter without establishing cross-patient typing. Low fine-subtype scores alone cannot distinguish limited APT information from annotation ambiguity, imbalance, preprocessing, or model limitations. A useful benchmark therefore needs explicit statistical units, reference models, and error diagnostics, rather than interpreting a leaderboard as an assay information limit.

APT-Bench specifies this task with 361,792 cells from 40 participants, 293 APT measurements, and RNA-derived labels for five lineages and 27 subtypes. Five fixed patient-disjoint folds and subject-balanced pooled Macro-F1 define the primary comparison. Classical and neural recipes establish reference performance; Soft-cascade is one hierarchy-aware reference, not a separate algorithmic contribution. Split controls reveal protocol-dependent optimism, while paired-RNA reconstruction and patient-composition recovery contextualize the cell-level results. The contribution is this clinical APT task and its auditable interface, not the invention of paired profiling or subject-disjoint evaluation. Sampling analyses remain in a separate report.

The paper makes one contribution with three concrete components:

- **A specified evaluation contract:** APT-only lineage and subtype prediction with a fixed ontology, patient-disjoint folds, and explicit cell-weighted versus subject-balanced metrics.

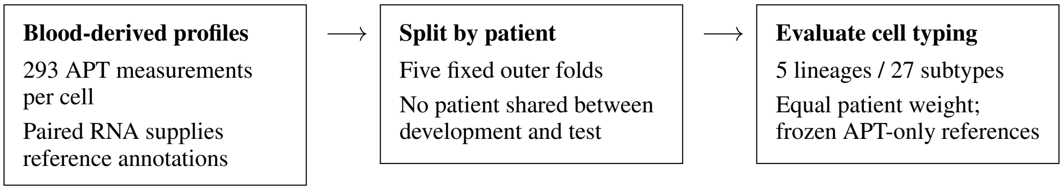

Figure 1: The primary APT-Bench contract. Patient-disjoint outer folds evaluate cell typing in new individuals; subject-normalized confusion pooling assigns equal total scoring weight to each patient. Cell-random splitting is a separate diagnostic, not a substitute for this generalization target.

- **A split-protocol audit:** cell-level split controls yield higher scores in both evaluated model families at both label resolutions, demonstrating why held-out-cell and new-patient performance must be distinguished.
- **Reference results and an auditable interface:** classical and neural predictions, metric code, split manifests, and documented artifact checks establish comparison points for subsequent APT models.

A complete public benchmark also requires accessible data, verified permissions, and portable reproduction. Those release requirements remain explicit where unfinished; a public source repository alone does not establish their completion. The study does not claim validated blood-based diagnosis, a universal sampling law, or priority over all prior aptamer evaluation resources.

2 Related Work

Aptamer profiling and paired measurements. Apt-seq combines aptamer surface-binding measurements with transcriptomes using shared cellular barcodes (Delley et al., 2018). APT-Bench builds on that concept rather than claiming its invention. Its focus is a clinical PBMC reference task with a fixed APT panel, RNA-derived labels, and patient-disjoint evaluation. The paired-RNA probe asks how these annotations are reconstructed under matched training budgets, not whether joint profiling itself is novel.

Single-cell representations and hierarchy. Geneformer, scGPT, and scFoundation motivate large transcriptomic representations, while controlled studies show that simple baselines can remain competitive (Theodoris et al., 2023; Cui et al., 2024; Hao et al., 2024; Kedzierska et al., 2025). Cell annotation also motivates hierarchical losses and architectures (Jia et al., 2025; Cultrera di Montesano et al., 2026). We use flat, HCE, and soft-cascade Transformers alongside classical models as APT references. Because their complete objectives are not matched, this ladder illustrates the evaluation setting rather than constituting a new learning algorithm or component ablation.

Evaluation across cells and subjects. Open Problems separates datasets, methods, and metrics (Luecken et al., 2025); multimodal benchmarks additionally compare input configurations (Liu et al., 2025). Neither a coarse/fine ontology nor subject-disjoint evaluation is individually a new principle. APT-Bench makes the statistical unit explicit for APT cell typing and contrasts new-patient evaluation with cell-level split controls. Gradient-boosted trees provide strong tabular references (Grinsztajn et al., 2022; Erickson et al., 2026). Separate sampling and feature-budget reports do not constitute additional primary contributions or resolve biological versus technical confounding.

3 Benchmark Setting

3.1 Blood Profiles and Reference Labels

APT-Bench is derived from peripheral-blood specimens from 40 participants in five cancer groups (CRC, GC, HCC, PC, and BTC) and healthy controls, with 6–8 participants per group. The paired assay measures transcriptomic state and 293 target-specific nucleic-acid aptamers on the same cells. Transcriptomic counts were processed with Seurat (Hao et al., 2021); predicted doublets and low-quality cells were removed, and clusters were manually annotated using canonical markers from prior PBMC and pan-cancer studies (McGinnis et al., 2019; Zhang et al., 2024; Yang et al., 2024). These annotations define five lineages and 27 subtypes. They are operational reference labels, not independently measured biological truth.

The linked post-QC cohort contains 374,854 cells. Excluding the uncertain *Unknown* subtype leaves 361,792 benchmark cells. Each subtype has one fixed parent lineage. Disease is a six-class patient attribute, not the root of the cell hierarchy: any given lineage can occur in multiple disease groups. The benchmark therefore connects cell and patient tasks through aggregation, rather than imposing a disease-to-lineage-to-subtype label tree.

The construction checklist and unresolved acquisition and annotation fields are recorded in Appendix B.1. In particular, the analyzed data do not establish a validated one-drop input-volume requirement. The scope here is characterization from the profiled blood cells, not end-to-end validation of a small-volume diagnostic assay.

3.2 Patient-Disjoint Cell-Typing Contract

The evaluation has a primary task and a secondary application. **Cell identity** predicts lineage and subtype using APT alone at inference. Training supervision is recipe-specific: only Soft-cascade additionally uses development-patient disease labels for contrastive pairing. **Patient characterization** predicts disease-group membership from patient summaries of measurements or cell-model outputs. The latter is an exploratory association task: with one acquisition per participant and incomplete acquisition metadata, disease biology and technical structure cannot be identified separately. Open-set subtype discovery is outside the contract; all primary cell tasks use the fixed five-lineage/27-subtype ontology.

Five frozen disease-stratified folds each hold out eight patients. All cells from a patient remain together; preprocessing, model selection, and fitting are restricted to development patients. We pool predictions across the five out-of-fold (OOF) partitions. A cell-random split is retained only as a leakage diagnostic, not as evidence of generalization to new patients. The release interface records cell and patient IDs, ontology, folds, predictions, and metrics (Appendix B.4). Full data access and a portable end-to-end evaluator remain release requirements.

3.3 Metrics and Statistical Units

Coarse and fine subject-balanced pooled Macro-F1 (SB-Macro-F1) are the primary cell metrics. For patient p, let $C^{(p)}$ be the confusion matrix over a fixed K-class ontology and n_p its number of evaluated cells. Define

$$\overline{C} = \frac{1}{P}\sum_{p=1}^{P}\frac{C^{(p)}}{n_p}, \qquad \text{SB-MacroF1} = \frac{1}{K}\sum_{k=1}^{K}\frac{2\overline{C}_{kk}}{\sum_j \overline{C}_{kj} + \sum_i \overline{C}_{ik}}. \tag{1}$$

A zero denominator contributes zero. Each patient's entire matrix, not each class row, is normalized to unit mass. This differs from cell-weighted pooled Macro-F1 and from mean within-subject Macro-F1. We retrospectively re-score the frozen OOF predictions under this primary reporting convention; checkpoints are not reselected. Historical cell-weighted scores remain in Table 15.

Main-table intervals use 2,000 paired patient-bootstrap resamples, with the same patient multiplicities across models. They are pointwise, conditional on the fitted models and fixed fold design, and do not include model-selection or training-seed uncertainty. The main comparison is descriptive; an interval for an individual score is not a test of pairwise superiority. Historical cell-weighted tests against HCE and XGBoost remain separately documented and are not transferred to the new metric. Exact-path accuracy, subtype-tree distance, root-excluded hierarchical F1, and consistency are appendix diagnostics.

Patient characterization uses one prediction per patient, with patient-level accuracy, macro-F1, and macro-AUROC. Its effective disease sample is 40, not 361,792. Nested model selection prevents direct test exposure but does not remove confounding or uncertainty from exploratory analysis choices.

The paired-RNA probe also uses subject-balanced pooled Macro-F1, but has a different training budget and preprocessing protocol; its scores must not be interpreted as additional architecture rows in the main table. Appendix B.9 gives the definitions.

3.4 Supporting Reliability Analyses

The paired-RNA analysis is an internal reconstruction diagnostic, since RNA also supplies the reference annotations. In the current matched-input probe, HVG selection is fitted separately within inner and outer training subsets; this prevents feature-selection exposure to their evaluation patients, but does not make RNA an independent validation of RNA-derived labels. Centering, patient-identity prediction, and recorded QC controls probe alternative explanations for disease association. Feature-panel and inference-cell subsampling are secondary redundancy diagnostics, not evidence of reduced training requirements or clinically efficient assays.

Subject–cell allocation experiments are preserved in a standalone supporting report (`sampling_report.tex`), not included as a contribution of the present paper. Their external RNA cohorts do not validate APT cell typing or APT disease prediction.

4 Benchmark Reference Models

APT-only comparisons. LR and XGBoost provide linear and nonlinear tabular references. Flat and HCE Transformers and a Soft-cascade Transformer provide backbone-matched neural recipes. All use APT at inference and the same outer patient partitions. Training and validation follow each recipe's documented protocol; identical outer folds do not imply identical training-set sizes or fully matched tuning. Linear SVM, Random Forest, and reference correlation remain in the full appendix inventory. A full-budget plain MLP/ResNet evaluation is not available; its absence is a coverage limitation, not evidence of incompatibility.

Hierarchy-aware reference. Soft-cascade Transformer embeds aptamer identities and values and combines a global subtype head with lineage-specific experts weighted by coarse probabilities. Appendix A specifies the routing equation and training objective. The independent coarse and fine heads are evaluated without constrained decoding. Flat removes the cascade; HCE instead uses a two-level hierarchical loss adaptation (Cultrera di Montesano et al., 2026). Only Soft-cascade uses cross-disease supervised contrastive pairing, so their objectives and auxiliary disease-label usage differ. These are reference-recipe comparisons, not component ablations. No routing or contrastive contribution is claimed in isolation. Appendix A gives the full architecture, losses, hyperparameters, and component matrix.

Input compatibility. Gene-based pretrained models such as scGPT and Geneformer (Cui et al., 2024; Theodoris et al., 2023) do not directly accept anonymous aptamer feature IDs without a verified mapping or a separately trained adapter. Supplying paired RNA would change the input contract. This distinguishes an undefined direct APT input from an adaptable but unevaluated model; it does not justify claiming superiority over transcriptomic annotation methods.

Secondary analyses. A separate matched-input probe varies APT and RNA while holding its training budget and patient folds fixed. The patient-level case study uses disease-label-free XGBoost cell predictions and nested cross-fitting: disease-training compositions come from inner held-out patients, and disease-test compositions come from the untouched outer fold. Downstream LR selection remains within development patients. These analyses contextualize APT typing; they are not additional primary benchmark tasks or validated clinical endpoints.

5 Results

5.1 Cell-Level Splitting Gives an Optimistic Evaluation

The benchmark first asks whether evaluation on additional cells represents generalization to additional patients. Table 1 contrasts the historical cell-level split controls with the frozen patient-disjoint evaluation. Cell-level splitting gives higher scores for both models at both resolutions: $+0.050/+0.069$ coarse/fine Macro-F1 for XGBoost and $+0.090/+0.108$ for Soft-cascade. The apparent fine advantage of Soft-cascade over XGBoost is also larger under cell splitting (0.048 versus approximately 0.009 under patient-disjoint evaluation).

Task	Model	Patient-disjoint	Cell split	Inflation
Coarse lineage	XGBoost	0.322	0.372	+0.050
	Soft-cascade Transformer	0.326	0.417	+0.090
Fine subtype	XGBoost	0.143	0.212	+0.069
	Soft-cascade Transformer	0.152	0.260	+0.108

Table 1: Historical cell-weighted pooled Macro-F1 under patient-disjoint evaluation versus the available cell-level split controls. All four model–resolution comparisons are higher under cell splitting. These are protocol comparisons, not matched-seed causal estimates of leakage alone; no uncertainty for the protocol difference is asserted. Differences were computed before rounding. The separate APT-only leaderboard (Table 2) reweights patients and therefore uses a different metric.

Model	Input	Coarse SB-Macro-F1 ↑	Fine SB-Macro-F1 ↑
Logistic Regression	APT	0.298 [0.281, 0.315]	0.126 [0.107, 0.139]
XGBoost	APT	0.325 [0.302, 0.345]	0.137 [0.120, 0.149]
Flat Transformer	APT	0.326 [0.300, 0.348]	0.139 [0.114, 0.155]
HCE Transformer	APT	0.335 [0.310, 0.356]	0.143 [0.121, 0.160]
Soft-cascade Transformer	APT	0.332 [0.306, 0.354]	0.145 [0.122, 0.160]

Table 2: Primary APT-only cell-typing comparison: five patient-disjoint folds, 40 patients, 361,792 cells. Entries are subject-balanced pooled Macro-F1 with pointwise 95% patient-bootstrap intervals (2,000 resamples). Frozen predictions are retrospectively re-scored with equal total patient weight; checkpoints are unchanged. Comparisons are descriptive, conditional on fitted models. Transformer auxiliary supervision differs (Table 3); full-budget MLP coverage remains unavailable.

These comparisons use the historical cell-weighted metric in both columns; they must not be compared numerically with the subject-balanced main leaderboard below. They contrast evaluation protocols, including their training/test allocation, rather than estimating a causal leakage effect under a fully matched randomized design. They support a specific conclusion: held-out-cell performance should not be presented as evidence of new-patient generalization. They do not show that all model rankings reverse or quantify optimism for every possible cohort.

5.2 Reference Performance Under the Patient-Disjoint Contract

Table 2 compares APT-only reference predictions using subject-balanced pooled Macro-F1 on the same 40 held-out patients. Coarse/fine scores are 0.298/0.126 for LR and 0.325/0.137 for XGBoost. Flat, HCE, and Soft-cascade Transformers reach 0.326/0.139, 0.335/0.143, and 0.332/0.145. These point estimates establish reference performance, not a statistically demonstrated cascade advantage.

The complete inventory and historical cell-weighted scores are in Tables 14 and 15. Fine prediction remains difficult across references. However, five-lineage and 27-subtype tasks differ in class count, prevalence, and annotation boundaries; their score difference is not an isolated assay-resolution effect. Full class-level and hierarchy diagnostics are in Appendix B.8. Missing full-budget MLP coverage and objective differences between Transformer recipes limit conclusions about architecture choice.

A privileged-label diagnostic masks each frozen fine score vector to the children of the true lineage, without retraining or changing routing. Fine SB-Macro-F1 increases to 0.366/0.361/0.358 for Flat/HCE/Soft-cascade (Appendix B.7). This measures the benefit of knowing the correct parent; it is not deployable performance or evidence that remaining errors reflect an intrinsic APT information limit.

5.3 Secondary Context: RNA and Patient Summaries

The paired-input diagnostic uses a capped budget, separately from the main leaderboard. MLP fine SB-Macro-F1 is 0.584 for RNA and 0.606 for RNA+APT (paired difference $+0.022$; 95% CI $[0.013, 0.037]$). RNA+APT also exceeds RNA+noise and RNA+within-patient-shuffled APT; the latter difference is $+0.010$ ($[0.002, 0.017]$). The corresponding SGD shuffled contrast is unresolved. These single-seed controls support a limited cell-pairing association; repeated shuffles and a wider RNA representation have not been evaluated. RNA supplies the annotations, so this is internal label reconstruction (Appendix B.2).

As a limited aggregation case study, nested subtype summaries reduce mean absolute reference-composition error from 0.02314 for a training-patient-mean baseline to 0.01478 (paired reduction 0.00836; 95% CI $[0.00648, 0.01010]$). This corresponds to mean total variation decreasing from 0.312 to 0.200, a rescaling of the same error rather than independent evidence. Lineage error instead changes from 0.05495 to 0.05827, with no resolved improvement. Appendix B.6 reports all classes and a direct median-APT ridge control (subtype MAE 0.01421). The soft-minus-direct difference is unresolved ($+0.00057$; $[-0.00092, 0.00211]$), so aggregation has no demonstrated advantage over direct patient regression. Disease association and recorded-QC conditional comparisons are confined to Appendix B.5.

6 Discussion

The central lesson of APT-Bench is that the unit of generalization must be part of the cell-typing benchmark. The observed cell-split optimism is consistent across the two evaluated model families and both resolutions, but its magnitude is specific to the available protocol comparisons. Patient-disjoint testing and subject-balanced scoring address different issues: the former separates people across training and test sets, while the latter limits unequal cell recovery from dominating the score.

A reusable comparison setting. The resource specifies an APT input, operational ontology, fixed patient folds, metrics, and reference results. Soft-cascade is a hierarchy-aware reference, not a claimed algorithmic advance. These references establish comparison points for improving APT cell typing without changing the input modality, ontology, or held-out patient partitions. Complete data access, permissions, annotation provenance, and a portable evaluator remain necessary release requirements. Full-budget MLP coverage and more comprehensive APT-compatible annotation baselines also remain limitations.

Boundaries of the evidence. The APT cohort contains 40 participants and RNA-derived reference labels. Independent annotation checks and additional APT cohorts are needed to assess broader generalization. Patient-disjoint testing does not remove disease-correlated acquisition effects, so the secondary disease study is not clinical validation. Paired RNA contextualizes label reconstruction rather than establishing an aptamer information limit. Subject–cell sampling analyses are preserved in a separate report; neither a scaling law nor a resource-allocation algorithm is a contribution of this paper.

7 Conclusion

APT-Bench establishes a patient-disjoint evaluation setting for cell typing from single-cell aptamer profiles, with a fixed label ontology, explicit subject-balanced scoring, and classical and neural reference results. Cell-level splitting produces higher scores in both evaluated model families at both resolutions, showing why held-out cells must not substitute for held-out patients in reporting generalization. The resulting comparison setting supports development of APT models against a clearly defined new-patient target. Complete reproducible release and independent APT cohorts are the next steps toward a broadly usable benchmark.

AI use statement

Generative-AI assistance through ChatGPT and Codex was used in the revision workflow for research-design suggestions, literature search, manuscript editing, analysis-code development, debugging, experiment orchestration, and scientific figure preparation. AI-generated suggestions were not themselves experimental evidence; reported computational results are tied to saved run artifacts. TODO: Authors: confirm the complete tool/model inventory and versions, describe the human checks actually performed on code, citations, figures, and statistical claims, and affirm responsibility for the final submission. Automated checks do not substitute for this author verification.

Ethics statement

APT-Bench is intended for research evaluation, not clinical decision-making. The small single-cohort sample and unresolved acquisition effects preclude claims of clinical validity or demographic fairness. Strong patient-specific signals also create a privacy concern: removing direct identifiers alone does not establish that cell profiles cannot be linked to their source individuals. Any data release must respect participant consent and applicable access terms. TODO: Data owners: supply the actual ethics approval or exemption and consent terms; confirm de-identification procedures, permitted sharing/access controls, and conflicts of interest. These facts have not been verified from the available analysis artifacts; preserve reviewer anonymity when providing documentation.

Reproducibility statement

The repository records the fixed five-fold partition in `benchmark/splits/patient_folds.json`, selected artifact checksums and structural audits in `benchmark/release_audit/`, and artifact-level reproduction commands in `REPRODUCIBILITY.md`. The latter distinguishes rebuilding figures/PDFs from rerunning models. Per-experiment protocol files record sampling settings and seeds; the controlled simulation includes per-replicate results and source hashes. A snapshot of the local analysis environment is provided, but is not a lockfile for all historical training runs. TODO: Finalize anonymous reviewer access, data permissions and licenses, archival versioning, original training environments, and a portable raw-data-to-results evaluator. The present repository is not yet a complete independently reproducible benchmark release.

Author Contributions

TODO: Add author contributions using CRediT roles (for example: conceptualization, methodology, software, validation, formal analysis, data curation, visualization, writing, supervision, and funding acquisition). Preserve anonymity in the review version if required.

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A Transformer Reference Implementation Details

This appendix gives the implementation details for the hierarchy-aware reference ladder introduced in Section 4: a flat Transformer, its HCE counterpart, and Soft-cascade Transformer. The cascade focuses on the cell-level hierarchy, lineage $\rightarrow$ subtype, while disease-association audits use separate tabular baselines and patient-level aggregation. The reference recipes differ in auxiliary supervision as well as hierarchy structure; their comparison does not isolate individual components.

A.1 Cell Encoder

Each cell is represented by a standardized APT vector $x \in \mathbb{R}^{293}$. We embed feature identity and feature value separately. For aptamer index i, we form a token

$$t_i = \mathrm{Emb}(i) + \mathrm{MLP}_{\mathrm{val}}(x_i), \qquad i = 1, \dots, 293. \tag{2}$$

A learnable `[CLS]` token is prepended to the resulting sequence and processed by a pre-norm Transformer encoder with 4 layers, 8 attention heads, hidden dimension 256, feed-forward dimension 512, GELU activations, and dropout 0.1. The final `[CLS]` representation

$$z = \mathrm{Encode}(x) \in \mathbb{R}^{256} \tag{3}$$

serves as the shared cell representation for downstream tasks.

A.2 Soft-Cascade Reference

Let $M = 5$ be the number of coarse lineages and $K = 27$ the number of fine subtypes. Each subtype k belongs to exactly one parent lineage $m(k)$. A coarse head produces lineage logits

$$c = W_c z \in \mathbb{R}^M, \tag{4}$$

with routing distribution

$$p(m \mid x) = \mathrm{softmax}(c/\tau)_m, \tag{5}$$

where $\tau = 1$.

Instead of a hard cascade, which would force subtype prediction to follow only the top predicted lineage, Soft-cascade Transformer uses a soft mixture of a global subtype path and lineage-specific experts:

$$s_k(x) = \underbrace{g_k(z)}_{\text{global path}} + \alpha\, p(m(k) \mid x)\, e_{m(k),k}(z) + \gamma \log\big(\epsilon + (1-\epsilon) p(m(k) \mid x)\big), \tag{6}$$

Here $g(z)$ is a global subtype head, $e_{m,k}(z)$ is the expert associated with lineage m, $\alpha = 1.0$, $\epsilon = 0.1$, and γ is ramped from 0 to 0.5 during early training. The global path retains a direct subtype-logit contribution alongside the routed expert contribution; we do not isolate its effect from the other components. Final predictions are

$$\hat{k} = \arg\max_k s_k(x), \qquad \hat{m} = \arg\max_m c_m. \tag{7}$$

A.3 Cross-Disease Contrastive Regularization

The reference model includes a cross-disease supervised contrastive term. Let

$$u = \frac{\mathrm{Proj}(z)}{\|\mathrm{Proj}(z)\|_2} \tag{8}$$

be a normalized projection. For anchor i, P_i contains same-subtype, different-disease cells, and N_i contains different-subtype cells. Self-pairs and same-subtype/same-disease pairs are excluded. The implementation uses positive probability mass, rather than the standard average of individual positive log-probabilities:

$$\mathcal{L}_{\mathrm{contrast}} = -\frac{1}{|V|} \sum_{i \in V} \log \frac{\sum_{j \in P_i} \exp(u_i^\top u_j / 0.1)}{\sum_{j \in P_i \cup N_i} w_{ij} \exp(u_i^\top u_j / 0.1)}, \tag{9}$$

where $V = \{i : |P_i| > 0\}$. Anchors without positives are excluded; a batch with no valid anchor returns zero loss. Although the code permits extra weight on same-lineage negatives, the inspected resolved configuration uses $w_{ij} = 1$ for every included pair. The contrastive multiplier is zero through epoch 5 and then $0.2 \min(1, (t-5)/10)$. This disease-supervised objective is not used by Flat or HCE and is not an independently validated mechanism.

A.4 Optimization and Model Scope

The classification cascade is trained jointly using

$$\mathcal{L} = \lambda_{\text{fine}}\mathcal{L}_{\text{fine}} + \lambda_{\text{coarse}}\mathcal{L}_{\text{coarse}} + w_{\text{contrast}}(t)\mathcal{L}_{\text{contrast}}, \tag{10}$$

where $\mathcal{L}_{\text{fine}}$ and $\mathcal{L}_{\text{coarse}}$ are cross-entropy losses with $\lambda_{\text{fine}} = 1.0$ and $\lambda_{\text{coarse}} = 0.5$. The contrastive weight increases to 0.2 according to the schedule in §A.3.

We optimize the cascade using AdamW with learning rate 10^{-4} and weight decay 10^{-5}, gradient clipping at norm 1.0, batch size 512, and mixed-precision training. Models are trained for at most 50 epochs with early stopping based on validation fine-level macro-F1 with patience 10. The final configuration does not use class-balanced sampling because it underperformed natural sampling after matching the number of optimization steps; the single-split development diagnostic is provided in Appendix B.11.

Backbone-matched Transformer references. The flat FT-style Transformer reference (Gorishniy et al., 2021) removes the cascade while retaining the identical APT tokenizer, Transformer dimensions, optimizer, early stopping, and training budget above. We call it FT-style because its tokenizer is exactly the APT feature tokenizer used by Soft-cascade Transformer, rather than the original numerical/categorical FT-Transformer tokenizer. This model and its HCE counterpart predict the same $M + K$ lineage-and-subtype nodes. The flat model applies independent cross-entropy to the lineage and subtype logit slices. For HCE (Cultrera di Montesano et al., 2026), let $R_{ij} = 1$ when node j is node i or its descendant, $p = \operatorname{softmax}(q)$ over all nodes, and $s = Rp$. We minimize $-\lambda_{\text{fine}} \log s_{M+k} - \lambda_{\text{coarse}} \log s_m$ for observed subtype k and lineage m. Because every fine label here is a leaf, the fine HCE term alone would reduce to ordinary CE; including the simultaneously observed parent label makes this a stated two-level adaptation rather than an unqualified reproduction of the mixed-granularity setting in the original work. Coarse HCE decoding maximizes subtree scores s_m, not raw internal-node logits; fine decoding maximizes leaf logits. All three Transformer models are evaluated from the checkpoint selected by patient-validation fine macro-F1, without an additional post-selection calibration stage. Table 3 makes the remaining objective differences explicit. In particular, only the soft-cascade reference uses disease labels to construct cross-disease contrastive pairs; Flat and HCE use no disease labels and no contrastive term. The ladder therefore holds backbone capacity, optimization budget, folds, and checkpoint selection, but it is not an objective-matched component ablation. We do not attribute any score difference to soft routing, hierarchical loss, or contrastive regularization in isolation.

Component	Flat FT-style Transformer	FT-style Transformer + HCE	Soft-cascade Transformer
APT tokenizer and Transformer dimensions	Same	Same	Same
Patient folds, optimizer, budget, checkpoint rule	Same	Same	Same
Fine-label supervision	Independent CE	Leaf HCE NLL	CE
Coarse-label supervision	Independent CE	Ancestor HCE NLL	CE
Cross-disease supervised contrastive term	No	No	Yes ($w \leq 0.2$)
Disease labels used by cell model	No	No	Contrastive pairing only
Global subtype path and lineage experts	No	No	Yes
Soft lineage routing	No	No	Yes

Table 3: Components of the three neural references. "Same" denotes an identical specification, not shared fitted weights. The models match the tokenizer, encoder dimensions, data folds, optimizer, maximum epochs, early stopping rule, and checkpoint criterion, but they do not match the complete training objective. In particular, disease labels enter only the soft-cascade contrastive pairing rule. Consequently, comparisons among these rows evaluate complete recipes and are not component ablations.

Disease recognition is not handled by a dedicated head inside Soft-cascade Transformer. The reference recipe addresses cell-level hierarchy prediction; disease-source prediction is evaluated separately as a within-cohort audit. High disease-source scores do not establish biological attribution or external clinical validity.

Strictly nested cell-output transfer audit. For each disease outer fold f, we train a disease-label-free XGBoost subtype classifier on all 32 outer-development patients and predict the cells of the eight untouched patients in f. Disease-training features require a second cross-fit: for each of the other four fixed folds g, a separate subtype model is trained on the 24 patients outside $f \cup g$ and predicts fold g. We average the resulting 27-class probability vectors within patient to obtain soft subtype composition; soft lineage composition sums subtype probabilities under the frozen subtype-to-lineage map. A standardized, class-weighted multinomial LR is fit to the 32 inner-OOF patient vectors, with $C \in \{0.01, 0.1, 1, 10, 100\}$ selected by patient-level inner CV, and evaluated once on the eight outer-test vectors. Raw proportions are primary; CLR with a prespecified 10^{-6} pseudo-count is a sensitivity analysis. All uncertainty and permutation analyses resample or shuffle patients, never cells. The outer-test fold is excluded from base fitting and downstream selection. Within the development set, however, meta-level selection reuses cached inner-OOF features: base models producing some meta-training features may have seen subtype labels from the meta-validation fold. This creates inner-selection cross-dependence, not exposure to the outer-test patients. A fully isolated meta-inner comparison would require another level of base refitting; it has not been performed. To test conditional rather than marginal subtype value, we additionally compare three nested feature sets under the same folds and inner selection: recorded-QC summaries alone, QC plus the five predicted lineage proportions, and QC plus the 27 predicted subtype proportions. The last representation contains lineage implicitly because child probabilities sum to their parents.

The main cell-identity results are reported in Table 2; the following sections supply their implementation and diagnostic details.

B Dataset and Evaluation Details

B.1 Dataset Construction and Documentation Checklist

The available study workflow establishes the high-level sequence used to construct APT-Bench: clinical peripheral-blood specimens from five cancer groups and healthy controls are converted to single-cell suspensions; cell-profile and viability quality control precedes a paired single-cell transcriptome and 293-aptamer assay; separate transcriptome and aptamer-barcode libraries are linked at the cell level; and transcriptome-derived clusters and annotations provide the cell labels. For a benchmark release, this overview must be accompanied by the following provenance and protocol fields.

Outstanding disease-confounding validation. TODO: Obtain a sample-linked acquisition manifest from the data owner: patient and sample IDs, collection and processing dates, APT assay batch, plate/pool, reagent lot, library-preparation batch, sequencing run, site, and available fresh/frozen and handling records. Audit disease-by-batch overlap before choosing a validation design. Where disease groups have adequate cross-batch support, evaluate patient-disjoint, held-out-batch generalization with all preprocessing and selection restricted to training data. If disease and batch are inseparable, obtain crossed-batch samples or an independent APT cohort; statistical adjustment alone cannot identify their separate contributions. The NC matrices must not be treated as external validation if they are the healthy subset of this cohort. These validations are pending, not completed. Expression and QC matrices alone do not establish acquisition provenance; current disease results remain within-cohort associations.

Transcriptomic preprocessing and cell quality control. The transcriptomic count matrix was processed with Seurat following its standard workflow of normalization, feature selection, dimensionality reduction, neighborhood-graph construction, and clustering (Hao et al., 2021). DoubletFinder was used to identify and remove predicted doublets (McGinnis et al., 2019). Low-quality cells were additionally filtered using per-cell UMI count, detected-gene count, and mitochondrial-transcript fraction. These steps preceded cluster annotation and construction of the linked APT benchmark cohort. TODO: Report the Seurat and DoubletFinder versions, every parameter and random seed, the exact UMI/gene/mitochondrial thresholds, and cell counts before and after each filter.

Manual annotation and reference-label reliability. After Seurat clustering, investigators manually assigned lineage and subtype names by inspecting canonical marker-gene expression reported in prior PBMC and pan-cancer studies (Zhang et al., 2024; Yang et al., 2024). No quantitative annotation confidence, inter-annotator agreement, or external reference-transfer score is available. Accordingly, these labels are operational reference annotations, not error-free biological ground truth, and benchmark scores are conditional on this manual taxonomy. The *Unknown* label was assigned when a cluster did not show a coherent canonical marker pattern for the expected PBMC subtypes; this was a manual reject decision rather than a calibrated confidence threshold. Unknown cells are excluded from the primary 27-subtype benchmark. TODO: Provide the full subtype-by-marker citation table, the number and expertise of annotators, whether they were blinded to disease, the adjudication procedure, and the explicit rule used to assign Unknown.

Cross-disease semantics and hierarchy mapping. The same 27 subtype labels and definitions were applied across all six disease groups; no subtype was renamed or redefined for a particular disease. This gives the labels a consistent operational meaning, but it does not prove that marker expression or biological state is invariant across diseases. Each subtype is assigned to one of five parent lineages by a manually curated lookup. The exported lookup and structural audit are available under `benchmark/release_audit/`: the existing subtype summary has 27 unique subtypes, five parents, and 361,792 cells, and the fold manifest has five disjoint groups of eight patients. These checks establish internal bookkeeping consistency, not independent biological validity. TODO: Document independent validation of the subtype-to-lineage mapping, including reviewers, disagreements, adjudication, and any marker-based checks.

B.2 Paired-Input Internal Reconstruction Reference

We use paired RNA as an internal label-reconstruction reference, separate from the primary APT-only model comparison. The current matched-input experiment replaces the earlier data-development HVG representation: 2,000 features are selected by mean-binned log-count dispersion within each inner or outer training subset, using the original RNA count matrix. All 361,792 benchmark cells are aligned by barcode. This controls feature-selection exposure to evaluation patients, but the transcriptomic modality also underlies the manual annotations. Stronger RNA reconstruction therefore does not independently validate those annotations or establish an intrinsic APT information limit.

Within each of the five fixed patient-disjoint folds, we compare five inputs: APT only, paired RNA only, APT+RNA, RNA plus random 293-dimensional noise, and RNA plus APT vectors shuffled within patient. The noise control checks feature dimension, while the within-patient shuffle preserves patient-level APT structure but breaks cell-level RNA–APT pairing. All inputs use the identical deterministic sample of up to 1,000 training cells per development patient and all cells from the eight untouched test patients. RNA features are selected as 2,000 highly variable genes using only the current training fold; all features are standardized using training-fold statistics. We fit patient- and class-weighted linear SGD and one-hidden-layer MLP probes for 40 epochs at both resolutions, using one inner patient-disjoint validation fold to select the regularization strength. Scores use subject-normalized OOF confusion matrices, and 2,000 paired patient-bootstrap replicates quantify modality contrasts. This matched-budget probe uses the main subject-balanced metric, but is not an architecture comparison with the full-training model ladder in Table 2.

Probe	Input	Coarse Macro-F1	Fine Macro-F1
SGD	APT only	0.277 [0.254, 0.296]	0.090 [0.076, 0.098]
SGD	RNA only	0.742 [0.731, 0.750]	0.431 [0.387, 0.453]
SGD	APT + RNA	0.751 [0.740, 0.762]	0.458 [0.413, 0.483]
SGD	RNA + noise	0.742 [0.732, 0.751]	0.409 [0.371, 0.430]
SGD	RNA + shuffled APT	0.747 [0.732, 0.760]	0.448 [0.405, 0.471]
MLP	APT only	0.310 [0.292, 0.325]	0.094 [0.081, 0.103]
MLP	RNA only	0.871 [0.854, 0.883]	0.584 [0.529, 0.610]
MLP	APT + RNA	0.873 [0.858, 0.884]	0.606 [0.551, 0.635]
MLP	RNA + noise	0.870 [0.855, 0.880]	0.571 [0.525, 0.593]
MLP	RNA + shuffled APT	0.876 [0.862, 0.887]	0.596 [0.544, 0.622]

Table 4: Paired-input internal reconstruction diagnostic under the five fixed patient-disjoint folds. Each input uses the same deterministic sample of up to 1,000 training cells per development patient, training-fold standardization, all held-out cells, patient- and class-weighted training, and one inner patient-disjoint validation fold for regularization. Brackets are 2,000 paired patient-bootstrap intervals for subject-balanced pooled Macro-F1. RNA features are selected separately within inner and outer training subsets. This prevents feature-selection exposure to their evaluation patients, but the RNA-derived operational annotations are not independently validated by RNA reconstruction.

For the MLP probe, paired RNA improves coarse/fine macro-F1 over APT by $+0.562$ and $+0.490$. The concatenated representation improves fine macro-F1 over RNA alone by $+0.022$ ($[0.013, 0.037]$), and exceeds RNA plus shuffled APT by $+0.010$ ($[0.002, 0.017]$). The linear SGD probe shows the same direction for RNA+APT versus RNA ($+0.028$; $[0.013, 0.041]$), but its shuffled-APT contrast is unresolved ($+0.010$; $[-0.005, 0.025]$). RNA therefore reconstructs the operational taxonomy far better than APT, while the small combined-input gain should be read as internal complementary association rather than an estimate of annotation accuracy or a clinical claim.

- TODO: Report recruiting site(s) and dates, prospective versus retrospective collection, exact patient count per group, inclusion/exclusion and diagnostic criteria, and whether each participant contributed exactly one specimen.
- TODO: Report available demographics and clinical covariates, including age, sex, disease stage, treatment status, and comorbidities; summarize missingness and state which fields cannot be released.
- TODO: Report blood volume, collection tubes, time to processing, cell-isolation and cryopreservation procedures, quality-control thresholds, and sample/cell exclusion counts.
- TODO: Report assay/platform and reagent versions, sequencing instruments and depth, 293-probe panel design, and a verified protein-target manifest or explicitly state why only anonymized probe identifiers can be released.
- TODO: Report barcode-matching rates, APT normalization and filtering, all remaining software versions and parameters, and whether annotation was performed before investigators examined APT profiles or disease labels.
- The documented final cell-flow accounting is 374,854 linked post-QC cells $-$ 13,062 Unknown cells $=$ 361,792 benchmark cells. TODO: Data owners: provide participant and cell counts at all earlier collection/QC exclusions before a complete cohort-flow diagram can be drawn. Supply the approval, consent, access, and licensing evidence listed in the Ethics and Reproducibility Statements.

B.3 Chance-Adjusted Resolution Diagnostic

Raw coarse and fine Macro-F1 compare five-class and 27-class tasks with different label frequencies and decision boundaries. We therefore treat their difference as descriptive, not evidence of a resolution deficit beyond task cardinality. Using the released 40 patient-specific OOF confusion matrices from the matched paired-input diagnostic, we normalize each matrix to unit mass and average over patients to obtain M. Let $p_k = \sum_j M_{kj}$ and $q_k = \sum_i M_{ik}$. The independence reference

$M_{ij}^0 = p_i q_j$ retains both true and predicted marginals while breaking their association. We report

$$S_0 = \frac{1}{K}\sum_k \frac{2p_k q_k}{p_k + q_k}, \qquad S_{\mathrm{adj}} = \frac{\mathrm{MacroF1}(M) - S_0}{1 - S_0}, \tag{11}$$

with zero contribution when $p_k + q_k = 0$. This is F1 of an independence matrix, not the exact expectation of finite-sample permutation F1. Each of 2,000 paired patient-bootstrap draws recomputes M, both marginals, and the adjustment; the same patient draw is used for both resolutions and all modalities. These intervals condition on the fitted models and do not include retraining variation.

Probe	Input	Resolution	Raw F1	Chance ref.	Adjusted F1 [95% interval]
SGD	APT	coarse	0.277	0.185	0.113 [0.093, 0.134]
SGD	APT	fine	0.090	0.033	0.059 [0.046, 0.067]
SGD	RNA	coarse	0.742	0.197	0.678 [0.666, 0.689]
SGD	RNA	fine	0.431	0.035	0.410 [0.367, 0.433]
SGD	APT + RNA	coarse	0.751	0.198	0.690 [0.676, 0.702]
SGD	APT + RNA	fine	0.458	0.035	0.438 [0.393, 0.463]
MLP	APT	coarse	0.310	0.186	0.152 [0.134, 0.171]
MLP	APT	fine	0.094	0.029	0.067 [0.055, 0.076]
MLP	RNA	coarse	0.871	0.200	0.839 [0.819, 0.854]
MLP	RNA	fine	0.584	0.037	0.568 [0.512, 0.596]
MLP	APT + RNA	coarse	0.873	0.200	0.842 [0.824, 0.855]
MLP	APT + RNA	fine	0.606	0.037	0.591 [0.535, 0.621]

Table 5: Marginal-chance adjustment for the paired-input reconstructability diagnostic. Scores use subject-balanced pooled confusion matrices from the same 40 OOF patients. Each of 2,000 paired patient-bootstrap draws recomputes the observed score and null marginals. This is an association-breaking reference, not a task-cardinality, annotation-quality, or within-lineage difficulty control.

For APT, the MLP raw coarse/fine scores 0.310/0.094 have chance references 0.186/0.029 and adjusted scores 0.152/0.067. The difference shrinks from 0.216 to 0.085 (95% interval $[0.068, 0.104]$). The corresponding SGD adjusted gap is 0.054 ($[0.034, 0.078]$). Adjusted RNA and APT+RNA differences remain larger for both probes, around 0.25–0.27. The adjustment accounts for marginal chance agreement, but does not control label boundaries, class count, annotation reliability, or within-lineage separability; none of these residual differences establishes a cardinality-independent biological granularity effect.

The separate frozen APT-only neural references additionally undergo true-lineage restricted decoding in Appendix B.7. It retains every held-out cell rather than selecting the easier subset whose coarse prediction is correct; it is not a conditional classifier trained anew.

B.4 Benchmark Interface and Release Contract

The primary hierarchy submission interface contains pseudonymous cell and patient IDs, true and predicted lineage, and true and predicted subtype. The disease interface contains one row per patient with the true disease and six class scores. The secondary budget diagnostic additionally records the aptamer measurement budget, inference-time cell budget, selection seed, and the development-only feature ranking. These common interfaces permit new models to be scored without reproducing their internal training pipelines.

The evaluation layer is portable to other clinical single-cell datasets. It implements group-disjoint out-of-fold prediction; patient-clustered bootstrap uncertainty; subject-balanced and cell-weighted pooled coarse and fine Macro-F1, exact-path accuracy, root-excluded hierarchical F1, subtype-tree distance, the descriptive coarse-minus-fine difference, and sibling versus cross-lineage error; patient-level aggregation versus cell-level pseudoreplication; patient-centering and identity diagnostics; and nested feature- and cell-budget curves. A future dataset can replace the assay features, hierarchy, or clinical endpoint while retaining these interfaces. This modular task–dataset–metric separation follows recent living single-cell benchmarks (Luecken et al., 2025); standardized metadata and programmatic access similarly enable cross-study reuse in CZ CELLxGENE (CZI Cell Science Program et al., 2025).

(a) Task and evaluation scope

Resource	Primary input	Reported scope	Main unit	Hierarchy
Open Problems	Multiple	12 living tasks	Task-specific	Task-specific
scMultiBench	RNA/ADT/ATAC	64 real + 22 simulated	Cell/dataset	No
SPARE	scRNA-seq	5 datasets, 8 methods	Sample	No
PaSCient	scRNA-seq	12.5M cells, 2,700 patients	Patient	No
APT-Bench	293 aptamers	1 cohort, 40 patients	Cell + patient	Yes

(b) Protocol components and availability

Resource	Patient task	Two-axis scale	Assay budget	Public pipeline
Open Problems	Task-specific	No	No	Yes
scMultiBench	No	No	Feature selection	Yes
SPARE	Yes	No	No	Yes
PaSCient	Yes	No	No	Not audited
APT-Bench	Secondary	Separate report	Secondary	Pending release

Table 6: Scope comparison with recent single-cell evaluation resources (Luecken et al., 2025; Liu et al., 2025; Shitov et al., 2025; Liu et al., 2026). "Task-specific" means that the living platform defines multiple contracts rather than one shared clinical protocol. APT-Bench is narrower in cohorts but centers a fixed aptamer modality and patient-disjoint coarse-to-fine cell typing. Patient-level analysis is secondary; sampling studies are in a separate report and do not expand the main contribution. Public release remains a submission requirement, not a completed claim.

Aggregation / Representation	Patient Acc.	Macro-F1	Macro-AUROC	Perm. p
Log cell count + LR‡	0.575	0.531	0.800	0.00010
All recorded QC proxies + LR¶	0.550	0.566	0.859	–
Nested soft subtype composition + LR§	0.600	0.593	0.854	0.00010
Nested soft lineage composition + LR§	0.475	0.477	0.766	0.00020
Reference subtype composition + LR‡	0.500	0.516	0.776	0.00010
Reference lineage composition + LR‡	0.325	0.301	0.743	0.0184

‡ Patient-level technical or paired-transcriptomic oracle features. ¶ Recovered cell count plus five patient summaries of recorded RNA UMI and detected-gene QC fields; see Appendix B.5. § Soft probabilities from disease-label-free XGBoost under strict nested cross-fitting. Composition rows use raw proportions; LR regularization is tuned only within each outer-development set. Composition permutation tests use 10,000 patient-label shuffles.

Table 7: **Associational patient-characterization audit** ($n = 40$). Predicted composition retains disease-associated variation, but its gains over lineage composition, recovered cell count, and recorded RNA-QC proxies have patient-bootstrap intervals spanning zero. These representations may contain biological, patient, or technical structure. Full direct-APT settings and confounding controls appear in Appendix B.5; no row establishes biological mediation or clinical generalization.

A complete release must contain the processed APT matrix; pseudonymous cell and patient identifiers; disease and cell labels; hierarchy and feature metadata; the immutable fold manifest; metric implementations; baseline configurations; and a single command that regenerates the headline tables from model predictions. The fold manifest, exported hierarchy, selected checksums, and artifact-rebuild instructions are now recorded in `benchmark/release_audit/` and `REPRODUCIBILITY.md`; these cover selected results, not the complete dataset or training pipeline. TODO: Before submission, finalize anonymous data/code access, explicit licenses, original training environment locks, an archival DOI, and the end-to-end command. If participant-level data require controlled access, release the complete schema, all labels permitted by consent, the split manifest, evaluator, and access procedure.

B.5 Confounding Audit of Within-Cohort Disease Association

Exploratory scope and independent sample size. All disease analyses reuse one cohort of 40 independent patients across six groups (6–8 patients per group); the 361,792 cells are repeated observations, not additional independent disease outcomes. Outer-fold disease models are developed on only 32 patients. Nested cross-fitting separates training from evaluation but does not increase the number of independent subjects, remove unmeasured confounding, or turn repeated analyses into independent replications. Regularization and inner validation mitigate model-selection risk without eliminating the uncertainty associated with this small development sample. The representation comparisons and post-hoc sensitivities below are exploratory, not confirmatory evidence of incremental subtype value or clinical utility. Reported patient-bootstrap intervals do not account for the full adaptive history of analysis choices. Clinical validation would require an independent APT cohort with recorded acquisition covariates and a frozen preprocessing, feature-selection, and prediction pipeline; the external scaling experiments do not supply that disease-validation cohort.

Table 7 summarizes the patient-representation comparison within this appendix.

The multi-model cell-level disease ranking is archived, not a patient-level endpoint. The representative centering and patient-ID diagnostics below retain the evidence of patient-associated signal.

Non-identifiability of disease and acquisition effects. Every participant contributes one acquisition, so patient identity is nested within both disease group and any unrecorded processing batch. The released metadata contain disease and patient identifiers plus transcriptomic UMI and detected-gene counts, but no plate, run, date, library, operator, or site field. Consequently, neither patient-disjoint prediction nor label permutation can identify whether a patient-associated APT offset is biological, technical, or mixed. No analysis below is interpreted as disease biology, biomarker validation, or clinical performance.

Recorded nuisance-only baselines. We aggregate the only available per-cell QC fields to each patient: recovered cell count; medians of $\log(1 + \mathrm{UMI})$ and $\log(1 + \mathrm{detected\ genes})$; median detected-gene/UMI ratio; and the interquartile ranges of the two log-count variables. Each feature set uses the same outer folds as the composition bridge, four-fold inner patient CV on the 32 outer-development patients to select LR regularization, refitting on all 32, and one evaluation on the sealed eight patients. Table 8 shows that these recorded QC proxies alone approach the predicted-composition result. They remain proxies rather than measured batch labels, so this analysis detects known nuisance structure but cannot rule out unrecorded acquisition effects.

We also test conditional subtype value rather than comparing each representation in isolation. Under the same nested folds and inner regularization selection, recorded QC plus predicted lineage reaches macro-F1 0.583 and AUROC 0.875; recorded QC plus the full predicted subtype vector reaches 0.599 and 0.825. The subtype-minus-lineage macro-F1 increment is $+0.016$ with 95% paired patient-bootstrap CI $[-0.120, 0.169]$ (10,000 resamples). Thus subtype detail does not provide a resolved conditional increment in this cohort.

Recorded feature set	d	**Acc.**	**Macro-F1**	**Macro-AUROC**
Recovered cell count	1	0.575	0.531	0.800
RNA QC summaries	3	0.475	0.478	0.787
All recorded QC proxies	6	0.550	0.566	0.859
Predicted soft subtype composition	27	0.600	0.593	0.854
Recorded QC + predicted lineage	11	0.575	0.583	0.875
Recorded QC + predicted subtype	33	0.600	0.599	0.825

Table 8: Matched patient-level nuisance audit. All rows use the same outer folds and inner patient-level regularization selection. The predicted-composition minus all-recorded-QC Macro-F1 difference is $+0.027$ with a 95% paired patient-bootstrap interval of $[-0.157, 0.215]$ (2,000 resamples). Recorded QC fields are incomplete proxies, not batch labels. In the conditional audit, the subtype-minus-lineage Macro-F1 increment is $+0.016$ with 95% paired patient-bootstrap interval $[-0.120, 0.169]$ (10,000 resamples).

Patient-wise centering sensitivity. To test whether disease prediction relies on each patient's absolute expression level rather than only on within-patient relative structure, we repeated Logistic Regression training and evaluation after subtracting each patient's own per-aptamer median from every cell in both training and test data. This transformation preserves within-patient cell-to-cell variation while removing a patient-specific location shift. Cell-level accuracy drops from 0.991 to 0.401 (macro-F1 $0.989 \rightarrow 0.348$), and patient-level majority-vote accuracy drops from $40/40$ to $26/40$. This collapse shows that disease recognition depends substantially on patient-level absolute-expression structure, but it does not by itself determine whether that structure is biological, technical, or a mixture of both.

Patient-fingerprint diagnostic. We additionally trained a 40-way classifier to predict patient identity from individual cells, using a within-patient 50/50 cell split for this diagnostic only. The classifier identifies the correct patient from a single held-out cell with 88.4% accuracy (macro-F1 0.862), against a 2.5% chance level. Individual cells therefore carry a strong and easily recoverable patient-specific signature, consistent with the centering sensitivity above.

Patient-level summary-feature control. Finally, to confirm that the disease signal does not rely only on cell-level pseudo-replication, we trained a genuinely patient-level classifier using each patient's median 293-dimensional APT profile. This summary-feature model reaches patient accuracy 0.950, patient macro-F1 0.948, and macro-AUROC 1.000. Together with the sensitivity analyses above, this indicates that disease information is present at the patient level but strongly entangled with absolute-expression offsets that cannot be cleanly separated without acquisition-batch metadata.

Strict nested cell-output transfer audit. Ordinary pooled OOF predictions are insufficient for stacking because a base model used to create a disease-training feature may have seen the eventual disease outer-test patients. We therefore use the nested protocol defined in Methods: within each outer fold, four XGBoost models trained on 24 patients generate inner-OOF features for the 32 disease-training patients, and a fifth model trained on all 32 predicts the eight outer-test patients. XGBoost receives subtype labels but no disease labels. Soft subtype composition reaches accuracy/macro-F1/AUROC $0.600/0.593/0.854$ and soft lineage composition reaches $0.475/0.477/0.766$. The paired macro-F1 difference is $+0.116$ (95% patient-bootstrap CI $[-0.015, 0.262]$). Gold subtype and lineage compositions reach macro-F1 0.516 and 0.301, respectively. The apparently stronger predicted than gold point estimate is not statistically resolved and warns that predicted probabilities may encode model-confidence or technical structure beyond true cell abundance.

This experiment is therefore a representation-transfer audit, not an error- propagation analysis. It does not establish that the transferred signal originates from subtype abundance, that fine subtype is more useful than coarse lineage, that better typing improves patient characterization, or that the signal is independent of confidence and technical structure. Consistent with this distinction, the soft-minus-hard subtype macro-F1 difference is only $+0.069$ (95% patient-bootstrap CI $[-0.077, 0.221]$), and the predicted-minus-gold subtype difference is $+0.077$ (95% CI $[-0.100, 0.256]$).

Across 10,000 patient-label permutations, predicted subtype and lineage composition have plus-one $p = 0.00010$ and $p = 0.00020$. However, log cell count alone reaches macro-F1 0.531 ($p = 0.00010$), and the predicted-subtype minus cell-count difference is $+0.062$ (95% CI $[-0.087, 0.218]$). Equal-cell subsampling at 100, 500, and 2,000 cells per patient preserves the qualitative result over 50 seeds. A 50-repeat global-multinomial composition control averages macro-F1 0.170, and fold-ID prediction from global OOF composition is below chance. The permutation code reruns downstream LR selection for each relabeling; the disease-label-free cell models remain fixed. However, the original folds were constructed using disease strata and are kept fixed. Unconditional exchangeability given this design has not been established, so the reported tail fractions are exploratory randomization diagnostics, not confirmatory biological tests. These controls support a predictive and associational bridge, not causal mediation or independent biological validation.

Method	Lineage MAE	Lineage TV	Subtype MAE	Subtype TV
Training-patient mean	0.05495	0.137	0.02314	0.312
Soft cell-output mean	0.05827	0.146	0.01478	0.200
Median APT ridge	0.05324	0.133	0.01421	0.192

Table 9: Patient-composition recovery on the same 40 outer-test patients. Lower is better. Mean TV equals $K\,\mathrm{MAE}/2$ and is not independent evidence. Direct regression uses inner-selected ridge and simplex projection; soft composition uses frozen nested XGBoost predictions. Soft-minus-direct MAE differences are $+0.00503$ ($[-0.00415, 0.01469]$) for lineage and $+0.00057$ ($[-0.00092, 0.00211]$) for subtype (pointwise paired 95% patient-bootstrap intervals). Neither contrast establishes equivalence.

B.6 Patient Composition Recovery

For each of the 40 outer-test patients, we compare the mean soft cell-type probabilities with reference cell fractions. The constant baseline is the equal-patient mean reference composition of the 32 outer-development patients in that fold; no test-patient annotations enter its construction. For K classes and N test patients, the recovery error is

$$E = \frac{1}{N}\sum_{p=1}^{N}\frac{1}{K}\sum_{k=1}^{K}|\widehat{\pi}_{pk} - \pi_{pk}|.$$

We report $E_{\mathrm{baseline}} - E_{\mathrm{model}}$, so positive values favor predicted compositions. All five lineages or all 27 subtypes are included, without selecting well-recovered classes. Absolute errors across these two resolutions are not directly comparable because their denominators differ. Using 2,000 disease-stratified paired patient-bootstrap replicates, subtype error decreases from 0.02314 to 0.01478, a reduction of 0.00836 ($[0.00648, 0.01010]$). Lineage error changes from 0.05495 to 0.05827, a reduction of -0.00332 ($[-0.01389, 0.00670]$). Intervals are pointwise and conditional on the completed cross-fitted cell models; they do not include training-subset or model-fitting uncertainty. The accompanying per-class audit reports error, signed bias, and rank correlation for every class. Disease-adjusted rank correlations residualize both ranked fractions on disease-group indicators and are descriptive, not evidence of biological specificity or causal independence from disease. Reference recovery does not validate the annotation itself or imply improved disease characterization.

Total variation and direct regression. For probability vectors, $\mathrm{TV}(\widehat{\pi}, \pi) = \frac{1}{2}\sum_k |\widehat{\pi}_k - \pi_k|$, so mean TV is exactly $KE/2$. For subtypes, baseline/model mean TV is 0.312/0.200; this is the same information as MAE, not a second validation endpoint. Per-class bias and centered recovery errors include every class, including rare subtypes.

We additionally fit patient-median APT directly to reference composition using multioutput ridge regression. Each outer fold uses the remaining four frozen patient folds to select $\alpha \in \{0.1, 1, 10, 100, 1000\}$ by MAE after Euclidean projection onto the probability simplex. Standardization is fitted within each training split. This control uses provided APT expression values, not a newly selected normalization, and no disease labels enter fitting. Table 9 reports all three methods. The soft-minus-direct subtype MAE difference is $+0.00057$ ($[-0.00092, 0.00211]$), so there is no resolved advantage of cell-output aggregation over direct regression. The new intervals use 2,000 disease-stratified paired patient draws with seed 20260912; their Monte Carlo endpoints differ slightly from the earlier audit. All point estimates for the existing soft and constant methods reproduce their saved results. Full per-class MAE, bias, and centered-error outputs are provided with the analysis, without selecting favorable classes.

Patient-label permutation control. This is a disease-classification diagnostic, not a composition-recovery test. Because the frozen folds were originally disease-stratified, unrestricted label permutations need not preserve the original fold-construction mechanism; the following tail fraction is conditional on the retained protocol rather than a design-exact test of biological disease specificity. We evaluated the same patient-level median-profile classifier under 1,000 label permutations. Disease labels were permuted across the 40 patients as indivisible units, preserving the disease-group counts $(6, 8, 7, 7, 6, 6)$; labels were never shuffled across cells. We retained the original five patient folds rather than restratifying after permutation. No permuted dataset reached the observed accuracy of 0.950 or macro-F1 of 0.948, giving plus-one permutation $p = (0 + 1)/(1000 + 1) = 0.000999$ for both statistics. The null accuracy mean was 0.137, with empirical 95% interval $[0.025, 0.275]$. This rejects a random patient-label explanation under the fixed protocol, but does not distinguish biological signal from patient-associated technical structure.

B.7 True-Lineage Restricted Decoding

We replay the 15 frozen final checkpoints (three neural references, five patient-disjoint folds), retain explicit cell IDs and ontology columns, and require exact reconstruction of their original coarse and fine predictions before computing this diagnostic. For each of the same 361,792 held-out cells, we mask the fine score vector to subtypes whose parent is the true lineage, then take its argmax. We do not change Soft-cascade routing or condition the evaluation set on a correct coarse prediction. The diagnostic therefore uses privileged test-label information and cannot be placed on the deployable leaderboard. Table 10 reports subject-balanced pooled Macro-F1 and paired patient-bootstrap differences, conditional on these frozen models. Reduced candidate-set size contributes to the gain; neither the gain nor the remaining error identifies an assay-specific information ceiling.

Model	Ordinary fine	True-lineage oracle	Paired gain [95% CI]
Flat Transformer	0.139	0.366	+0.227 [0.202, 0.238]
HCE Transformer	0.143	0.361	+0.218 [0.196, 0.231]
Soft-cascade Transformer	0.145	0.358	+0.214 [0.194, 0.230]

Table 10: Privileged true-lineage restricted decoding on the same frozen predictions. Scores are subject-balanced pooled Macro-F1; gains use 2,000 paired patient-bootstrap resamples and pointwise intervals, conditional on fitted models. The oracle knows the true parent and is not a deployable method.

B.8 Per-Subtype Support and Performance Audit

Figure 2 visualizes the two marginal support–F1 associations for Soft-cascade Transformer and XGBoost. The vertical stack at 40 patients makes the coverage ceiling explicit. These plots and the partial correlations are exploratory: they neither establish causality nor test whether two dependent correlations differ significantly.

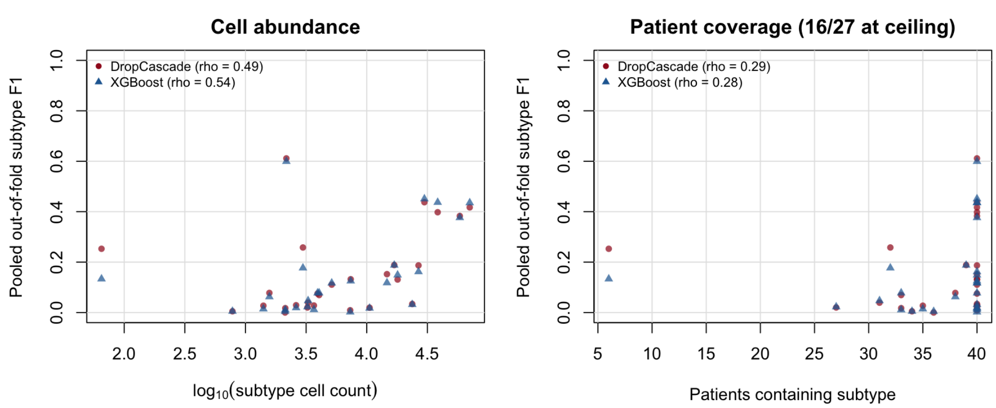

Figure 2: Per-subtype pooled out-of-fold F1 versus cell abundance (left) and patient coverage (right). Each marker is one of 27 subtypes; both models use the same ground-truth support values. Spearman correlations are marginal, and 16 of 27 subtypes occur in all 40 patients.

(a) Support–performance associations

Model	ρ_c	ρ_p	$\rho_{c\|p}$	$\rho_{p\|c}$
Soft-cascade Transformer	0.486	0.288	0.411	-0.051
Flat FT-style Transformer	0.505	0.255	0.464	-0.123
FT-style Transformer + HCE	0.513	0.318	0.427	-0.034
XGBoost	0.540	0.275	0.496	-0.130
Logistic Regression	0.435	0.107	0.488	-0.267
Linear SVM	0.339	-0.012	0.463	-0.335
Random Forest	0.661	0.418	0.564	-0.035
Reference Correlation	0.376	0.003	0.499	-0.354

(b) Granularity and error structure

Model	G	R_{sibling}	**Cross-lineage**
Soft-cascade Transformer	0.174	0.250	0.750
Flat FT-style Transformer	0.176	0.243	0.757
FT-style Transformer + HCE	0.183	0.257	0.743
XGBoost	0.179	0.246	0.754
Logistic Regression	0.167	0.295	0.705
Linear SVM	0.200	0.265	0.735
Random Forest	0.158	0.191	0.809
Reference Correlation	0.099	0.225	0.775

Table 11: Exploratory subtype diagnostics on cell-weighted pooled patient-disjoint predictions. ρ_c and ρ_p are marginal Spearman associations of subtype F1 with log cell count and patient coverage; partial correlations residualize ranked variables. $G = F_1^{\text{coarse}} - F_1^{\text{fine}}$. Among errors, R_{sibling} retains the true parent and Cross-lineage is its complement. These cohort-level summaries are descriptive, not causal.

Table 12 gives all subtype-level quantities and 95% percentile intervals from 2,000 patient-clustered bootstrap resamples. Counts and coverage use pooled ground truth, while F1 uses the same patient-disjoint OOF predictions as Table 2, but uses cell-weighted per-class F1 rather than subject-balanced scoring.

Table 12: Complete subtype support and cell-weighted per-class OOF F1 (not SB-F1), with pointwise 95% patient-bootstrap intervals. Counts and patient coverage refer to the same 361,792 cells.

Subtype	Lineage	Cells	Patients	Soft-cascade	XGBoost
Erythrocyte	Other	65	6	0.253 [0.000, 0.395]	0.133 [0.000, 0.545]
Atypical Memory B	B	781	34	0.005 [0.000, 0.014]	0.005 [0.000, 0.014]
CD4 Monocyte-4	Myeloid	1,405	35	0.027 [0.006, 0.053]	0.014 [0.000, 0.033]
platelet	Other	1,574	38	0.078 [0.016, 0.125]	0.063 [0.015, 0.106]
Tcm	T	2,121	36	0.000 [0.000, 0.000]	0.004 [0.000, 0.016]
Memory B-2	B	2,130	33	0.017 [0.004, 0.033]	0.011 [0.000, 0.024]
Plasma	B	2,173	40	0.611 [0.522, 0.684]	0.599 [0.496, 0.685]
pDC	Myeloid	2,615	40	0.029 [0.010, 0.061]	0.019 [0.005, 0.040]
CD14 Monocyte-3	Myeloid	2,977	32	0.258 [0.170, 0.306]	0.177 [0.103, 0.211]
CD8 + CTL-3	T	3,244	27	0.020 [0.004, 0.035]	0.023 [0.008, 0.043]
GCB	B	3,283	31	0.039 [0.007, 0.071]	0.047 [0.001, 0.073]
Cycling T	T	3,667	40	0.028 [0.015, 0.043]	0.012 [0.003, 0.022]
CD14 Monocyte-2	Myeloid	3,931	40	0.075 [0.046, 0.102]	0.078 [0.056, 0.097]
CD8 + CTL-2	T	4,049	33	0.069 [0.017, 0.116]	0.078 [0.015, 0.113]
CD16 Monocyte-2	Myeloid	5,142	40	0.111 [0.064, 0.161]	0.117 [0.063, 0.164]
CD8 + CTL-1	T	7,303	40	0.009 [0.000, 0.024]	0.002 [0.001, 0.004]
cDC2	Myeloid	7,376	40	0.132 [0.099, 0.181]	0.125 [0.079, 0.167]
CD8 Tem-2	T	10,548	40	0.019 [0.005, 0.038]	0.017 [0.008, 0.030]

Subtype	Lineage	Cells	Patients	Soft-cascade	XGBoost
CD16 Monocyte-1	Myeloid	14,678	40	0.152 [0.111, 0.198]	0.118 [0.090, 0.150]
CD4 + Tcm-2	T	16,861	39	0.188 [0.111, 0.236]	0.188 [0.136, 0.220]
Naive T	T	17,960	40	0.131 [0.067, 0.183]	0.149 [0.073, 0.199]
Memory B-1	B	23,722	40	0.034 [0.020, 0.048]	0.032 [0.017, 0.047]
CD8 Tem-1	T	26,714	40	0.187 [0.128, 0.250]	0.162 [0.107, 0.221]
CD14 Monocyte-1	Myeloid	29,871	40	0.437 [0.378, 0.495]	0.451 [0.392, 0.507]
CD56 NK	NK	38,442	40	0.398 [0.320, 0.436]	0.437 [0.373, 0.480]
CD16 NK	NK	58,507	40	0.383 [0.326, 0.434]	0.376 [0.317, 0.429]
CD4 + Tcm-1	T	70,653	40	0.417 [0.368, 0.460]	0.435 [0.393, 0.473]

Abundance does not alone explain the errors: Plasma has 2,173 cells and Soft-cascade CW-F1 0.611, whereas Memory B-1 has 23,722 cells and F1 0.034, and CD8+CTL-1 has 7,303 cells and F1 0.009. These examples complement the complete table; they do not establish the cause of subtype separability.

Cross-lineage error relative to an error-conditioned null. Table 13 uses the wrong-conditioned null as the primary reference: after excluding the true subtype, it renormalizes the predicted-subtype marginal so every randomized outcome remains an error. For the four main models, observed cross-lineage rates are 0.743–0.757 versus wrong-conditioned expectations of 0.741–0.745. The corresponding differences are only +0.002 to +0.013, and all four patient-bootstrap 95% intervals include zero. Thus the raw approximately 75% rate is largely explained by the available wrong-class distribution and does not independently demonstrate either hierarchy preservation or hierarchy failure. The unconditional permutation, which can create chance exact matches and therefore uses a different sample space, is retained only as a sensitivity analysis.

Model	Errors	Observed	Wrong-cond. null	Obs.−WC null [95% CI]	Uncond. null (sens.)
Soft-cascade Transformer	251,354	0.750	0.744	+0.005 [-0.025, +0.028]	0.675
Flat FT-style Transformer	247,798	0.757	0.745	+0.011 [-0.020, +0.035]	0.675
FT-style Transformer + HCE	250,944	0.743	0.741	+0.002 [-0.028, +0.023]	0.671
XGBoost	244,990	0.754	0.741	+0.013 [-0.020, +0.038]	0.672
Logistic Regression	301,566	0.705	0.752	-0.047 [-0.061, -0.036]	0.720
Linear SVM	296,904	0.735	0.774	-0.039 [-0.053, -0.028]	0.738
Random Forest	246,420	0.809	0.749	+0.059 [+0.021, +0.083]	0.685
Reference Correlation	328,596	0.775	0.785	-0.009 [-0.022, +0.003]	0.758

Table 13: Cross-lineage error relative to an error-conditioned marginal null. The primary wrong-conditioned (WC) null draws from each model's predicted-subtype marginal after excluding the true subtype and renormalizing, so every randomized outcome remains wrong. Observed−WC-null confidence intervals recompute both terms in 2,000 patient-clustered bootstrap resamples. The unconditional permutation preserves the realized true- and predicted-subtype marginals but can create chance exact matches; it is reported only as a sensitivity analysis.

B.9 Protocol Sensitivity: Patient-Level versus Cell-Level Splitting

The main-text Table 1 compares patient-disjoint and cell-level split protocols. This section defines their scoring units; the contrast does not isolate leakage from all training-allocation changes.

Model	Input	Coarse SB-Macro-F1 ↑	Fine SB-Macro-F1 ↑
Soft-cascade Transformer	APT	0.332 [0.306, 0.354]	0.145 [0.122, 0.160]
Flat Transformer	APT	0.326 [0.300, 0.348]	0.139 [0.114, 0.155]
HCE Transformer	APT	0.335 [0.310, 0.356]	0.143 [0.121, 0.160]
XGBoost	APT	0.325 [0.302, 0.345]	0.137 [0.120, 0.149]
Linear SVM	APT	0.309 [0.292, 0.324]	0.105 [0.089, 0.118]
Logistic Regression	APT	0.298 [0.281, 0.315]	0.126 [0.107, 0.139]
Random Forest	APT	0.253 [0.227, 0.277]	0.089 [0.079, 0.097]
Reference Correlation	APT	0.171 [0.147, 0.192]	0.068 [0.050, 0.078]

Table 14: Complete subject-balanced pooled Macro-F1 reference inventory, including auxiliary classical baselines. Definitions, resampling, and scope match Table 2.

Model	Input	Coarse CW-Macro-F1 ↑	Fine CW-Macro-F1 ↑
Soft-cascade Transformer	APT	0.326 [0.300, 0.350]	0.152 [0.129, 0.166]
Flat Transformer	APT	0.323 [0.299, 0.344]	0.147 [0.120, 0.161]
HCE Transformer	APT	0.333 [0.310, 0.354]	0.151 [0.127, 0.166]
XGBoost	APT	0.322 [0.298, 0.343]	0.143 [0.127, 0.158]
Linear SVM	APT	0.311 [0.296, 0.325]	0.111 [0.093, 0.126]
Logistic Regression	APT	0.299 [0.283, 0.315]	0.132 [0.113, 0.146]
Random Forest	APT	0.250 [0.228, 0.271]	0.092 [0.082, 0.100]
Reference Correlation	APT	0.172 [0.149, 0.189]	0.072 [0.055, 0.082]

Table 15: Historical cell-weighted pooled Macro-F1, with pointwise 95% patient-bootstrap intervals, for the frozen APT-only hierarchy predictions. The same predictions are reweighted in Table 2; the estimands differ. Classical auxiliary references are retained here rather than omitted from the benchmark.

Metric definitions. Let $C^{(p)} \in \mathbb{R}^{K\times K}$ be the confusion matrix for held-out subject p and $n_p = \sum_{ij} C_{ij}^{(p)}$. We distinguish

$$C^{\mathrm{CW}} = \sum_p C^{(p)}, \qquad F_1^{\mathrm{CW}} = \mathrm{MacroF1}(C^{\mathrm{CW}}), \tag{12}$$

$$C^{\mathrm{SB}} = \frac{1}{P}\sum_p \frac{C^{(p)}}{n_p}, \qquad F_1^{\mathrm{SB}} = \mathrm{MacroF1}(C^{\mathrm{SB}}), \tag{13}$$

$$F_1^{\mathrm{within}} = \frac{1}{P}\sum_p \mathrm{MacroF1}(C^{(p)}). \tag{14}$$

We call these *cell-weighted pooled Macro-F1*, *subject-balanced pooled Macro-F1*, and *mean within-subject Macro-F1*. Subject balancing normalizes each subject confusion matrix to unit mass before pooling; it is neither equal-cell subsampling nor averaging per-subject Macro-F1. Released files retain the legacy column name `patient_balanced_macro_f1` for F_1^{SB}, but no different patient-balanced statistic is used.

The numerical split comparison appears in the main results (Table 1). Its historical scores are cell-weighted in both columns, rather than the subject-balanced leaderboard metric.

Exact subject-normalized pooling is primary. The older equal-cell Monte Carlo sensitivity remains archived, rather than occupying a separate table.

B.10 Coarse–Fine Consistency Diagnostic

Independent coarse and fine classifiers can disagree about the hierarchy even when each prediction is plausible in isolation. Table 16 reports their natural agreement on the same frozen OOF predictions as the main results. These auxiliary path metrics remain cell-weighted averages, whereas the main Macro-F1 table now balances patient contributions.

The main reporting metric was revised retrospectively without retraining, retuning, or reselecting checkpoints. The historical cell-weighted comparison family contrasted Soft-cascade with HCE and XGBoost at coarse and fine resolution, using 2,000 paired patient bootstrap resamples and Holm correction over four tests; none reached adjusted significance. Those tests do not apply to the subject-balanced results. Flat and auxiliary classical comparisons remain descriptive. The saved re-evaluation script reproduces the old point estimates before generating the new table, and records source-artifact hashes.

Model	**Fine Acc.**	Root-excluded Hier. F1	**Consistency**	Exact Path	Subtype Tree Dist.
Soft-cascade Transformer	0.305	0.404	0.812	0.253	2.431
Flat FT-style Transformer	0.315	0.411	0.795	0.257	2.407
FT-style Transformer + HCE	0.306	0.405	0.834	0.259	2.418
XGBoost	0.323	0.420	0.736	0.248	2.375

Table 16: Cell-weighted auxiliary hierarchy diagnostics on the same frozen OOF predictions as Table 2. Root-excluded hierarchical F1 is the example-averaged F1 between true and predicted $\{\text{lineage}, \text{subtype}\}$ sets; the always-correct root is omitted. Exact Path requires both independent predictions to be correct. Consistency compares the mapped fine parent with the independent coarse prediction. Subtype Tree Distance uses only the predicted subtype and its mapped parent: 0/2/4 for correct/sibling/cross-lineage predictions (lower is better). *Unknown* is excluded; no constrained decoding is used.

B.11 Archived Supporting Analyses

Feature-panel diagnostics, version comparisons, and equal-cell sensitivity are indexed in the repository pruning manifest; original artifacts are retained.

Development-only sampling choice. A development-only class-balanced-sampling variant did not outperform natural sampling after matching optimization steps and was not retained.

B.12 Exploratory Leave-One-Disease-Out Stress Test

For each leave-one-disease-out (LODO) run, all patients from one group are withheld from training. The available table contains five cancer-group summaries. A numerical Normal-held-out artifact and row-level support counts have not been verified, so the sixth row is explicitly unavailable rather than omitted. These cell-weighted group summaries are not averaged or subtracted from pooled five-fold scores, since their evaluation populations and training protocols differ. This is an incomplete descriptive stress test; cross-disease generalization requires external, patient-clustered validation. Unmatched detector rankings are omitted.

Held-out Disease	**Fine Macro-F1**	**Fine Acc.**	**Coarse Macro-F1**	**Coarse Acc.**	**Tail Macro-F1**
BTC	0.086	0.317	0.310	0.479	0.073
CRC	0.105	0.277	0.309	0.520	0.112
GC	0.063	0.269	0.281	0.488	0.057
HCC	0.063	0.246	0.264	0.383	0.072
PC	0.057	0.253	0.221	0.482	0.037
Normal	—	—	—	—	—

Table 17: Available cell-weighted LODO summaries for Soft-cascade Transformer. Each numerical row withholds all patients of that disease. Dashes indicate that the numerical Normal-held-out artifact has not been verified, not zero performance. Per-row patient/cell/class supports are also unavailable in this summary; no six-group aggregate or comparison with pooled five-fold scores is made. A same-protocol simple baseline remains unevaluated.